

Ansh Bhansali

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EDUCATION

University of Illinois Urbana-Champaign, IL, USA

Aug 2025 – Dec 2026

Master of Engineering in Autonomy and Robotics | GPA: 3.9/4.0

Coursework: Humanoids, Autonomous Systems, Deep Learning, Computer Vision, Robot Planning, Control Systems

Sardar Patel Institute of Technology, Mumbai, India

Dec 2021 – Jul 2025

Bachelor of Technology, Electronics and Telecommunication | GPA: 3.9/4.0

EXPERIENCE

Research Engineer - IHSI, University of Illinois, IL, USA

Apr 2026 – Present

Project: ARPA-H MarginDx, AI-Guided Robotic Microsurgery for Tumor Margin Assessment

- Developed real-time **impedance force control** stack for **KUKA LBR Med 7 + PI hexapod** achieving tissue contact regulation.
- Built multi-layer IPC architecture (**Java/Python/UDP**) bridging KUKA, PC, and hexapod, with **sub-10ms** command latency.
- Achieved **10 μ m** resolution 3D tissue surface reconstruction with stable contact force during raster scans on tissue phantoms.

Embedded Engineering Intern - Dimension Six Technologies, Mumbai, IN

May 2024 – May 2025

Project: STM32-based E-Bike Conversion Kit

- Engineered production-ready firmware with battery management and regenerative braking, extending vehicle range by **40%**.
- Designed a 4-layer PCB in KiCad for a custom **ESC**, optimizing power stage layout to reduce losses by **15%** under peak load.
- Deployed IoT stack on **ESP32S3** with RFID authentication, enabling remote monitoring and contactless payments over MQTT.

Robotics Research Intern - Indian Institute of Technology, Bombay, Mumbai, IN

Jan 2024 – Jun 2024

Project: Autonomous Rugged Robot for Military Applications

- Built **SLAM**-based autonomous navigation stack with path planning, achieving **95%** accuracy in dynamic indoor environments.
- Fused IMU, GPS, and RGB-D camera data via **Extended Kalman Filter**, improving localization accuracy by **20%**.
- Deployed **YOLOv3** for real-time human detection and designed an adaptive gait controller for robust stair-climbing traversal.

SKILLS

- Languages:** Python, C++, MATLAB, Java, Bash
- AI & Perception:** VLA Models, Vision Transformers, PyTorch, RL (PPO), Computer Vision (YOLO, SAM), 6D Pose Estimation
- Planning & Control:** Motion Planning (A*, MPC), Trajectory Optimization (ProxQP, CHOMP), SLAM, Sensor Fusion, CBFs
- Tools & Frameworks:** ROS2, MuJoCo, Gazebo, Isaac Sim, Pinocchio, MoveIt, Git, Linux

PROJECTS

O-VLA Trajectory Optimization Safety Layer (ProxQP, CHOMP, RNEA, Pinocchio) ([Github](#))

- Built **morphology-agnostic** trajectory optimizer transforming unsafe VLA outputs into collision-free, constraint-satisfying paths.
- Achieved **100%** constraint satisfaction and **60-100%** collision reduction across 5 robot morphologies (6-23 DOF) in **0.03-0.44s**.
- Demonstrated **34-45%** energy reduction through coupled dynamics optimization using **Recursive Newton-Euler Algorithm**.

Humanoid Whole-Body Motion Planning (RL, MPC, Unitree G1, MuJoCo) ([Github](#))

- Implemented motion planning stack using **ZMP** preview control, **A*** footstep planning, and **MPC** balance optimization
- Achieved **49%** lower energy use than PD control with MPC balance control, while remaining stable under **70-80N** perturbations.
- Integrated **RL**-based locomotion with Jacobian IK manipulation for coordinated walk-and-reach tasks, achieving **75%+** success.

VLM-Based Autonomous Navigation System (GEM e4, GPT-4o, ROS2) ([Github](#))

- Built autonomous navigation stack integrating **GPT-4o VLM** for intent-based planning with LiDAR-camera sensor fusion.
- Implemented full pipeline: BEV perception, A* path planning, and Stanley control achieving end-to-end task success.
- Achieved **88%** accuracy on KITTI real-world dataset for VLM-based goal localization from natural language commands.

Open-Vocabulary 6D Pose Estimation (FoundationPose, SAM-3, Moondream2) ([Report](#))

- Developed **zero-shot** 6D tracking architecture with VLM-based **semantic inventory** and Gemini API query enrichment.
- Integrated **SAM-3** prompt segmentation with **Objaverse** mesh retrieval, achieving **76.5-100%** ADD-S AUC on YCBV dataset.
- Built RGB-D variant using **LangSAM** with **InstantMesh** and median-volume ensemble selection for mesh consistency.